

one to do, feel free to visit openProcessing.org—a large repository of user-uploaded Processing ‘sketches’.

Processing has simplified the way we look at code. Instead of source files, we call our programs ‘sketches’. A collection of our sketches is called a ‘sketchbook’. Since it’s built atop Java, it is platform independent, allowing us to develop for Windows, Linux and Mac.

Without further ado, let’s get started. The first thing to do is set up the environment. Once that is done, I will explain the basics of the Processing code syntax. Let us understand a sample program and run it in the Android emulator. I will then show you how to build and sign the APK (the app package) so that you can upload it to the market!

Setting up the environment

There are two parts to this: the first is to install the Android SDK and the second is to install Processing. First, make sure that the Java Development Kit (JDK) is installed and working. We will be using Linux as the base system for the purpose of this article (I will mention Windows now and again, wherever necessary, but the default assumptions will be from the Debian, Ubuntu and Linux perspective). If the JDK is not installed, you can download it using Synaptic or do so manually from Oracle’s website (links for all references including this one are given at the end of the article). For proper functioning of the Android SDK, use either `openjdk` or Oracle’s JDK—both would work.

A traditional Android SDK set-up involves the Eclipse IDE, but we will skip that bit (including the Eclipse ADT plugin installation) since our Processing IDE requires only a specific set of SDK binaries.

After installing the JDK, download the tarball—the file with the TGZ extension—from the Android developer website. Windows users can download the installer executable. Extract the tarball by executing the following commands in the terminal (from the directory where the tarball is located; alternatively, you can extract it using a GUI, i.e., File Roller or any other tool of your choice):

```
tar -zxvf android-sdk*.tgz
```

Once the files are extracted, change into the `/tools` directory within the SDK that we just extracted:

```
cd android-sdk-linux/tools
```

Then run the following command to invoke the Android SDK Manager:

```
android
```

In the rare case that *android* doesn’t work for you, try doing a `chmod` first—`chmod 777 android`, followed by `‘android’` or `./android`.

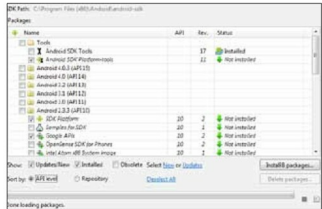


Figure 1: The Android SDK Manager

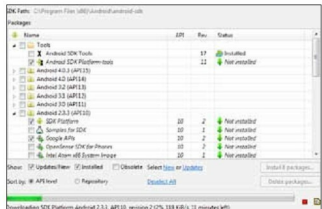


Figure 2: The SDK Manager is downloading and installing the selections we made

The SDK Manager will allow you to install the required SDK binaries and packages. Make sure to select the following packages (Note: The API number is very specific—it has to be API10, otherwise Processing may fail to execute our sketches on the emulator):

1. Tools > Android SDK Platform Tools (SDK Tools should already be installed; if not, select them as well)
2. Android 2.3.3 (API10) > SDK Platform
3. Android 2.3.3(API10) > Google APIs
4. Extras > Google USB Driver

Hit the *Install Packages* button and the Manager will then begin downloading and installing the SDK components. Once the process completes, we’re done with setting up the Android SDK bit of the environment.

We will now go on to install Processing, which is the easier part. Download a copy of the latest alpha version of Processing from the website. Yes, you read that correctly; we are living on the edge! At the time of writing this article, version 2.0 alpha 5 (2.0a5) is the latest. Note: Versions prior to this are known not to work for Android development due to the fact that Google has made several key changes to the SDK.

The Processing download will again be a tarball. Extract it just as we did for the SDK: